

N-PORT Filing Pipeline & Runbook

How the tool works, what the audit found, and what to do next — step by step.

Period	2026-06 (quarter ending 30 June 2026)
Funds	105 in custodian · 104 building · 1 cash-only
Status	All output marked TEST — nothing filed
Branch	humanreview-live-gate
Prepared	30 July 2026

Before you start

Both source extracts stop on **24 June**, but the filing covers through **30 June**. Nothing downstream can be trusted until they are re-pulled. That is Step 1.

1 · How the pipeline works

Three files in, one XML per fund out. Four commands, plus one trip to a Bloomberg terminal.

YOU SUPPLY

- **Custodian file** — every position, all funds (US Bank) → `data/custodian/`
- **AP order book** — every create / redeem order → `data/orders/`
- **EagleSTAR export** — the accountant's daily snapshots → `data/fund_accounting/`



1

nport masters
COMMAND

Reads all three inputs and builds **two Excel workbooks** covering every fund at once — one for holdings, one for fund-level numbers. Also writes the **provenance** and **reconciliation** reports.



2

Bloomberg terminal
MANUAL

Open both workbooks, let the `=BDP()` formulas calculate, save as `.xlsx`. This fills company IDs, country, returns and bond durations.



3

**nport
mergehumanreview**
COMMAND ×2

Lists whatever has no feed in a review spreadsheet. You fill the blanks; running it again folds your answers back in, tagged as **human-reviewed**.



4

nport split
COMMAND

Pure copying. Chops the workbooks into per-fund files. No decisions made here — which is why you **edit the workbook, never the fund files**.



5

nport build
COMMAND

Sorts each row into a type, validates every required field, checks the LIVE gate, writes the XML and validates it against the SEC schema.



YOU GET

- `output/<TICKER>_2026-06.xml` — one per fund, ready for EDGAR

The rule the design follows

Every number has exactly **one owner**. A second source never writes — it only cross-checks into the reconciliation report. Where nothing has a feed, a human fills it in and it gets tagged.

WHAT	WHO OWNS IT
Positions, share counts	Custodian
Company IDs, country, returns, durations	Bloomberg
Dollars — net assets, gains, flows	EagleSTAR accounting
Anything with no feed	Human review spreadsheet

2 · What the audit found

Every number here is measured by `scripts/audit_findings.py` from the files on disk. Re-run it after new data and this section updates itself.

Root cause — both source extracts stop short of period end

The filing covers through **30 June 2026**. The EagleSTAR trial balance ends **24 June** (latest of 121 snapshots) and the AP order book ends **24 June** (429 orders). Both are **6 days short**.

Reproduce: last file in `data/fund_accounting/.cache/tb/`; max parsed Trade Date in `HistoricalOrders (83).csv`. Parse the date — a string sort ranks 6/9 above 6/24.

What that does to the reconciliation

MEASURE	VALUE
Cross-check rows written	1,121
Flagged REVIEW	258
Funds with at least one flag	137 of 152
Funds currently clear to file LIVE	15

Most of those flags are harmless

Accounting books a trade when it *settles*; the order book records it when it is *placed*. A trade near a month boundary therefore lands in different months in the two systems. Summed over the quarter they agree exactly: **232 of 304** fund×side series tie to the cent. Only **72** genuinely differ.

Reproduce: sum mon1–3 per fund and side in `reconciliation_2026-06.csv` and compare the two source columns.

16 funds file a redemption of zero when money genuinely went out

Their redemptions were accepted on **6/24/2026 (16)** — the same day the accounting snapshot was cut, so they had not settled and never entered the trial balance. Because accounting is the *writer* for flows, the filings report **0.00**. Total unreported: **\$4,885,686**.

BRZX, HULL, MGKX, NYNY, OWN, STYL, WX, XAGI, XBIX, XCOM, XIWC, XKRE, XLEX, XLIX, XLVX, XVUG

Reproduce: quarter-sum redemptions per fund; look for `accounting = 0` with a non-zero order book.

Differences that survive the whole quarter

These are not timing. They need explaining before filing.

FUND	FLOW	ACCOUNTING	ORDER BOOK	DIFFERENCE
FDRS	Sales	112,593,659	81,424,626	+31,169,033
FDRX	Sales	19,604,246	10,133,360	+9,470,886
FDRS	Redemption	12,712,768	6,555,741	+6,157,027

EUV	Sales	462,028,250	464,905,260	-2,877,010
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Two code defects — found and fixed

DEFECT	BEFORE	NOW
Percentages did not tie to the reported total. Each holding's pctVal came from the custodian's net assets while the filing reports the accountant's — two denominators.	85 / 104	104 / 104
Swaps never stated long or short. The field was hardcoded blank though the direction is decoded from the ticker two lines earlier.	0 / 58	58 / 58

The percentage rule is now enforced on every build: a mismatch fails the build instead of shipping. Reproduce: compare `sum(pctVal)` with `sum(valUSD)/netAssets` in any output XML.

Known gaps that remain

ITEM	SCALE	ASSESSMENT
Option delta blank	70 of 70	No feed exists. FLEX options do not price on Bloomberg — from your risk system
Reinvestment blank	all funds	No feed exists. Not in an order book — needs the transfer agent
No designated index	7 funds	Fill in the review workbook if the prospectus names one
Real fund with no filing built	1	GPTZ — holds only a cash seed, no securities to report. Fails loudly, not silently. Confirm with compliance whether a cash-only series still owes a filing

Counted against the 105 accounts in the custodian file, not the 108 directories under `data/funds/` — 3 of those are hand-made test fixtures (`BOND_FUND`, `BUFFERED ETF`, `LEVERAGED ETF`) and are not fundable.

Checked and found sound

Recording these so the absence of a finding is not mistaken for an absence of checking.

CHECK	RESULT
SEC schema (XSD) validation	104 / 104 valid
Structure and content verifier	104 / 104 clean
Test suite	656 passing
Blank monthly returns — a gap, or genuine pre-inception?	104 / 104 track fund inception — not a gap
Every custodian account has a fund directory	all 105 accounts — none silently dropped
Filing status	104 TEST — nothing filed yet

3 · What to do — step by step

Work down the list. Do not skip Step 3 — everything after it inherits the answer.

1 Get a new EagleSTAR export through 30 June

Ask for it to be re-cut with a 30 June cutoff. Save the zip into:

```
data/fund_accounting/
```

2 Get a new AP order book through 30 June

Re-export it and save as:

```
data/orders/2026-06_orders.csv
```

The custodian file is already period-end. Leave it alone.

3 Check the dates actually moved

```
nport masters --dry-run
```

If either file still stops at 24 June, **stop here**. Everything downstream inherits the problem.

4 Build the workbooks

```
nport masters
```

Look for **flows from EagleSTAR TB + EagleSTAR gains/liabilities**. If it says *no fund accounting*, the zip was not found — stop and fix that first.

5 Read the reconciliation report

```
data/master/reconciliation_2026-06.csv
```

Most warnings should now be gone. Three things to look for:

- Warnings that cancel out across two months → ignore, that is settlement timing
- A fund with order-book redemptions but zero in accounting → investigate
- **FDRS** and **FDRX** → had real \$31M / \$9.5M gaps unrelated to dates

6 Go to the Bloomberg machine

Open both workbooks, wait for them to finish calculating, then save and close Excel.

```
data/master/security_master.xlsx  
data/master/filing_master.xlsx
```

Keep them as **.xlsx** — do not "save as CSV".

7 Create the review spreadsheet

```
nport mergehumanreview
```

8 Fill in the blanks

```
data/humanreview/2026-06_review.xlsx
```

Mainly the option deltas for the 18 defined-outcome funds. Save and close.

9 Merge your answers back in

```
nport mergehumanreview
```

Yes — the same command twice. First run creates it, second run absorbs your edits.

10 Push out to the funds

```
nport split
```

11 Test the build, then build for real

```
nport build --dry-run    # writes nothing, just reports problems
nport build
```

Expect **Built 104/108**. GPTZ failing is correct — it holds only cash.

12 Check the output

```
python scripts/verify_output.py
```

You want **CLEAN 104/104**. Then open two or three XMLs in `output/` and eyeball the fund name, period and net assets.

13 Go live

Set `liveTestFlag` to `LIVE` in `filing_master.xlsx`, save, then:

```
nport split
nport build
```

14 Upload the XMLs to EDGAR

If the build refuses to go LIVE

That is the safety gate doing its job — it blocks any fund whose reconciliation still has an unresolved warning. **Do not switch it off.** If it blocks nearly everything, the gate needs retuning to ignore the harmless month-shuffle warnings; ask before overriding.

Right now

Do Steps 1 and 2. Nothing else can move until those two files arrive.

4 · Where every data point comes from

Generated from the code by `scripts/field_provenance.py`, not written by hand. Every row cites the file and line that sets it, and the script refuses to run if a claim no longer matches the code.

Custodian Bloomberg EagleSTAR Human review Manual (workbook) fund_config.txt Tool constant
Derived

One writer per field

Each row below has exactly one source. A second system may *check* a value — the AP order book cross-checks EagleSTAR's flows, the custodian cross-checks net assets — but a cross-check never writes the filing. Disagreements go to `reconciliation_2026-06.csv` and block a LIVE filing until resolved.

FIELD	SOURCE	HOW IT IS OBTAINED	CODE REFERENCE
FUND IDENTITY			
cik	fund_config.txt	Typed once when the fund launches	src/nport/config.py:209
seriesId	fund_config.txt	From the SEC series guide spreadsheet	src/nport/config.py:209
classId	fund_config.txt	From the SEC series guide spreadsheet	src/nport/config.py:209
seriesName	fund_config.txt	Typed once when the fund launches	src/nport/config.py:209
seriesLei	fund_config.txt	Typed once (GLEIF-issued)	src/nport/config.py:209
regName / regCik / regLei	fund_config.txt	Trust-level, same for every fund	src/nport/config.py:209
regStreet1 ... regZip / regPhone	fund_config.txt	Trust address, set once	src/nport/config.py:209
signerName / signerTitle / signerOrg	fund_config.txt	Who signs the filing	src/nport/config.py:209
HOLDING IDENTITY			
name	Custodian	SecurityName, truncated to 30 chars	src/nport/custodian.py:245
title	Custodian	SecurityName verbatim	src/nport/custodian.py:247
cusip	Custodian	Copied via =custodian! cell reference; foreign CINS → N/A	src/nport/custodian.py:671
ticker	Custodian	StockTicker column	src/nport/custodian.py:226
isin	Bloomberg	=BDP(bbgid, "ID_ISIN")	src/nport/master_sheet.py:152
lei	Bloomberg	=BDP(bbgid, "LEGAL_ENTITY_IDENTIFIER"); N/A if the issuer has none	src/nport/master_sheet.py:153

invCountry	Bloomberg	=BDP(bbgid, "CNTRY_OF_DOMICILE") — no default, a blank is a real gap	src/nport/master_sheet.py:154
POSITION			
balance	Custodian	Shares / par from the custodian	src/nport/custodian.py:676
units	Tool constant	Set by holding type: NS shares, PA par, NC contracts	src/nport/custodian.py:677
curCd	Tool constant	USD — the book currency for every fund	src/nport/custodian.py:641
valUSD	Custodian	MarketValue. Swaps override to the EagleSTAR mark-to-market	src/nport/custodian.py:642
valUSD (swaps)	EagleSTAR	PVal Total Unreal G/L Base — the MTM, never the notional	src/nport/custodian.py:1051
pctVal	Derived	valUSD ÷ the netAssets the filing reports (not the custodian's)	src/nport/custodian.py:1053
payoffProfile	Custodian	Sign of the share count; for swaps the ticker's L/S code	src/nport/custodian.py:725
assetCat	Tool constant	Set by classification: EC / DBT / DE / STIV	src/nport/custodian.py:493
issuerCat	Tool constant	Set by classification: CORP / UST / RF / OTHER	src/nport/custodian.py:679
fairValLevel	Tool constant	1 for listed, 2 for derivatives	src/nport/custodian.py:680
isRestrictedSec / isCashCollateral / isNonCashCollateral / isLoanByFund	Tool constant	N for every ETF holding	src/nport/custodian.py:645
DEBT (C.9)			
maturityDt	Bloomberg	=BDP(cusip Govt/Corp, "MATURITY")	src/nport/master_sheet.py:171
annualizedRt	Bloomberg	=BDP(cusip Govt/Corp, "CPN")	src/nport/master_sheet.py:172
couponKind	Bloomberg	=BDP(... "CPN_TYP"), mapped to the N-PORT enum	src/nport/master_sheet.py:173
isDefault / areIntrstPmntsInArRs / isPaidKind	Tool constant	N — no defaulted debt held	src/nport/custodian.py:760
OPTION			
putOrCall	Custodian	Parsed from the position name (trailing C / P)	src/nport/custodian.py:520
exercisePrice	Custodian	Parsed from the position name (strike)	src/nport/custodian.py:697
expDt	Custodian	Parsed from the position name (expiry)	src/nport/custodian.py:699
writtenOrPur	Custodian	Sign of the contract count	src/nport/custodian.py:700

shareNo	Tool constant	N/A — index options are on an index, not a share count	src/nport/custodian.py:695
delta	Manual (workbook) NO FEED	FLEX options do not price on Bloomberg. Typed into security_master.xlsx and preserved across rebuilds — from your risk system	src/nport/master_sheet.py:585
refIndexName / refIndexIdentifier	Human review	option_index sheet → applied at split	src/nport/master_sheet.py:786
counterpartyName / counterpartyLei	Tool constant	The Options Clearing Corporation — every listed/FLEX option clears through it	src/nport/custodian.py:91
SWAP			
swapFlag	Tool constant	Y for every TRS	src/nport/custodian.py:732
terminationDt	Custodian	Parsed from the swap ticker (...-TRS---...)	src/nport/custodian.py:733
notionalAmt	Custodian	MarketValue = shares × price — the contract size	src/nport/custodian.py:734
unrealizedAppr	EagleSTAR	PVal Total Unreal G/L Base, matched on Primary Asset ID	src/nport/eaglestar.py:231
counterpartyName / counterpartyLei	Human review	swap_counterparties sheet — ticker code → legal name + GLEIF LEI	src/nport/master_sheet.py:773
recFixedOrFloating / recDesc	Human review	swap_legs sheet — from the trade confirm	src/nport/master_sheet.py:758
pmntFloatingRtIndex / pmntFloatingRtSpread / pmntRateTenor / pmntRateUnit	Human review	swap_legs sheet — from the trade confirm	src/nport/master_sheet.py:758
refCusip	Custodian	Leading segment of the swap ticker	src/nport/custodian.py:740
refIsin / refTicker / refIssuerName / refIssueTitle	Bloomberg	=BDP(refCusip Equity, ...) on the reference security	src/nport/master_sheet.py:190
FUND FINANCIALS (B.1)			
totAssets	EagleSTAR	Trial Balance total assets	src/nport/eaglestar.py:307
totLiabs	EagleSTAR	Trial Balance total liabilities	src/nport/eaglestar.py:309
netAssets	EagleSTAR	Trial Balance net assets (overrides the custodian figure)	src/nport/eaglestar.py:311
amtPayOneYrOther	EagleSTAR	Trial Balance payables	src/nport/eaglestar.py:304
assetsAttrMiscSec / assetsInvested / delayDeliv / standByCommit / liquidPref	Tool constant	0 — a genuine zero for a plain ETF, not a placeholder	src/nport/filing_master.py:72

amtPayOneYr* / amtPayAftOneYr* (other lines)	Tool constant	0 — no borrowings or affiliate payables	src/nport/filing_master.py:72
RETURNS (B.5)			
rtn1 / rtn2 / rtn3	Bloomberg	=BDP total return over each reporting month; N/A before inception	src/nport/filing_master.py:45
GAINS (B.5)			
netRealizedGainMon1-3	EagleSTAR	Trial Balance month-end deltas	src/nport/eaglestar.py:300
netUnrealizedApprMon1-3	EagleSTAR	Trial Balance month-end deltas	src/nport/eaglestar.py:301
FLOWS (B.2)			
mon1-3Sales / mon1- 3Redemption	EagleSTAR	Trial Balance subscriptions/redemptions. AP order book cross-checks only	src/nport/eaglestar.py:337
mon1-3Reinvestment	Tool constant NO FEED	Not in an order book — needs the transfer agent	src/nport/filing_master.py:80
RISK (B.3)			
durAdj / spreadDur	Bloomberg	=BDP DUR_ADJ_MID / OAS_SPREAD_DUR_MID per debt holding	src/nport/filing_master.py:50
maturity / ratingSP	Bloomberg	=BDP MATURITY / RTG_SP, bucketed by tenor	src/nport/filing_master.py:50
INDEX (B.6)			
nameDesignatedIndex / indexIdentifier	Human review	designated_index sheet — from the prospectus	src/nport/humanreview.py:91
ADMIN			
submissionType / isFinalFiling / isNonCashCollateral	Tool constant	NPORT-P / N / N	src/nport/filing_master.py:62
liveTestFlag	Tool constant	TEST until you flip it; LIVE is gated on a clean reconciliation	src/nport/cli.py:878
repPdEnd / repPdDate	Derived	Last calendar day of the filing period	src/nport/custodian.py:805
dateSigned	Derived	Set from the period; override in the workbook	src/nport/filing_master.py:115

Verified at generation time

- 65 rows: code anchor located for each
- Bloomberg claims checked against BLOOMBERG_SPECS (10 cols) + risk sheet (4 cols)
- EagleSTAR claims checked against 17 writer keys
- Human-review claims checked against 15 review value columns
- Custodian claims checked against 67 transform-assigned keys
- filing_master HEADER: 45 columns, 0 unmapped
- Coverage measured over **1,996 holdings** across **104 funds** in period 2026-06